

Deep Learning for Audio

Lecture 12

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AI Masters

2026

Outline

1. Music generation before 2020: MIDI data, MuseNet, Jukebox
2. Diffusion models: Stable Audio, Riffusion, Diffsound, Noise2Music
3. Transformer + Codec models: MusicGen, MusicLM, MeLoDy
4. SOTA trends: YuE, HeartMuLa, DiffRhythm 2, ACE-Step, Agents for Music

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Why music generating is important?

Music > Music News

Spotify Cracks Down on AI, Removing 75 Million Tracks and Targeting Impersonators

A three-pronged effort including a new impersonation policy, an updated spam filter, and standardized AI disclosures in music credits

- ▶ Spotify is cracking down on AI music spam: removed 75M AI tracks
- ▶ Scheme 1: AI clones of popular songs attract real listeners and divert royalties
- ▶ Scheme 2: AI tracks are streamed by bots to farm payouts
- ▶ New measures: stronger spam filters, stricter imitation rules, AI-use metadata disclosure
- ▶ Legitimate AI creators are still allowed if they avoid abusive practices

Spotify Cracks Down on AI, Removing 75 Million Tracks and Targeting Impersonators, 2025

Why music generating is important?

Echoes of Humanity: Exploring the Perceived Humanness of AI Music

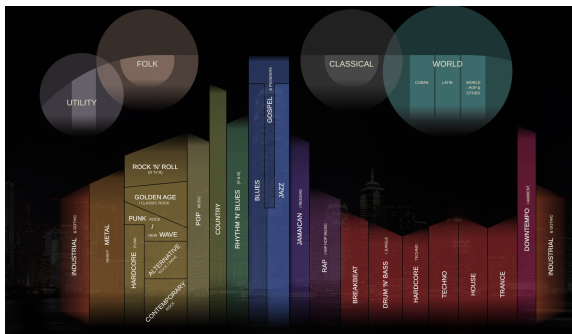
Flavio Figueiredo Giovanni Martinelli Henrique Sousa Pedro Rodrigues

Frederico Pedrosa Lucas N. Ferreira

Universidade Federal de Minas Gerais (UFMG)

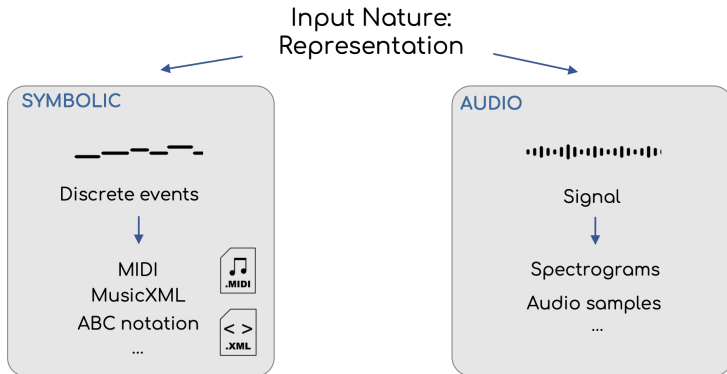
- ▶ In controlled comparisons, **AI-generated music is often perceived as human-made**
- ▶ Listeners tried to identify which track was AI-generated: one human-made, one AI-generated
- ▶ Detection accuracy dropped when tracks were closely matched in style and quality

Problems with generating music



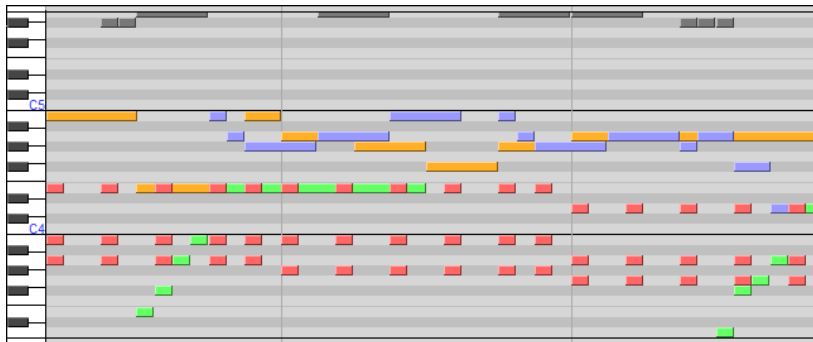
- ▶ Should NNs compose music by following the same logic and process as humans do?
- ▶ Are current evaluation methods good enough to compare and measure the creativity of the composed music?
- ▶ Difficult to collect data due to copyright
- ▶ Music is subjective
- ▶ Each music Genre has its own rules

Input representations



Basic Datasets

- ▶ **JSB Chorales Dataset** (chorales by Johann Sebastian Bach)
- ▶ **Maestro Dataset** (200 hours of virtuosic piano performances with fine alignment between note labels and audio waveforms)
- ▶ **The Lakh MIDI Dataset** (176,581 unique MIDI files)
- ▶ **MetaMIDI Dataset** (436,631 MIDI files)



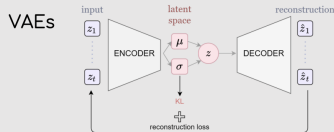
Overview of Music Datasets and Their Applications

Dataset Name	Year	Type	Scale	Main Application Areas
CAL500	2007	Audio	500 songs	Emotion Recognition
MagnaTagATune	2008	Audio	25,863 clips	Music Annotation, Emotion Recognition
Nottingham Music Dataset	2009	MIDI	1000 tunes	Symbolic Music Analysis
Million Song Dataset	2011	Audio	1,000,000 songs	Music Information Retrieval
MediaEval Emotion in Music	2013	Audio	1000 songs	Emotion Recognition
AMG1608	2015	Audio	1608 clips	Emotion Recognition
VCTK Corpus	2016	Audio	110 speakers	Speech Recognition, TTS
Lakh MIDI	2017	MIDI	176,581 files	Music Information Retrieval
NSynth	2017	Audio	300,000 samples	Music Synthesis
DEAM	2017	Audio	1802 songs	Emotion Recognition
LJSpeech	2017	Audio	13,100 clips	Speech Synthesis
Free Music Archive (FMA)	2017	Audio	106,574 songs	Music Classification
AudioSet	2017	Audio	2,000,000 clips	Audio Event Detection
CH818	2017	Audio	818 clips	Emotion Recognition
URMP	2018	Audio, Video, MIDI	44 performances	Audio-Visual Analysis
MAESTRO	2018	MIDI, Audio	200 hours	Music Generation, Piano Transcription
Groove MIDI Dataset	2019	MIDI, Audio	13.6 hours	Rhythm Generation
GiantMIDI-Piano	2020	MIDI	10,855 songs	Music Transcription, Analysis
LakhNES	2019	MIDI	775,000 multi-instrument examples	Music Generation
Slakh2100	2019	MIDI, Audio	2100 tracks	Source Separation
MG-VAE	2020	MIDI	2000 songs	Style Transfer
Groove2Groove	2020	Audio	thousands of clips	Style Transfer
Hi-Fi Singer	2021	Audio	11 hours	Singing Voice Synthesis
MIDI-DDSP	2022	MIDI, Audio	varied	Music Generation, Synthesis
Singing Voice Conversion	2023	Audio	subset of NHSS	Voice Conversion

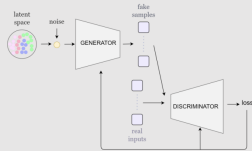
Chen et al., Applications and Advances of Artificial Intelligence in Music Generation: A Review, 2024

History: before 2019

Generative Models



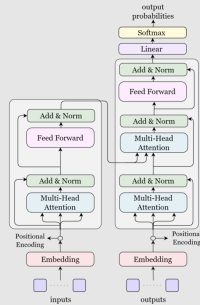
GANs



Sequence Models

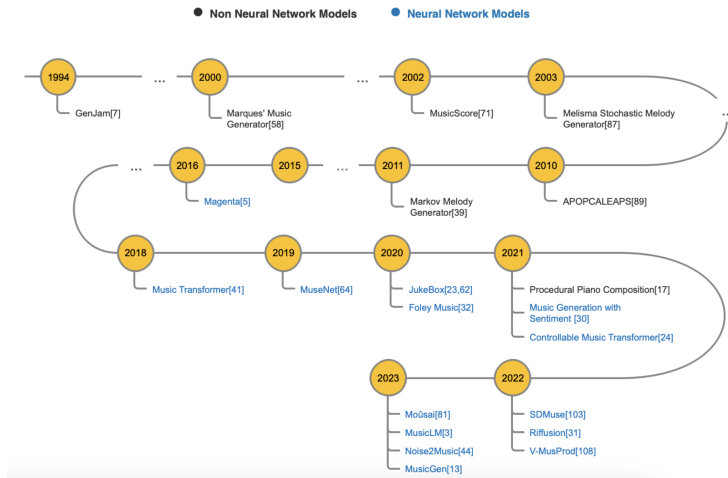
Recurrent Neural Networks (RNNs)

Transformers



Let's use models from CV/NLP and not give context about data domain (music)!

Music Generation Models: Timeline

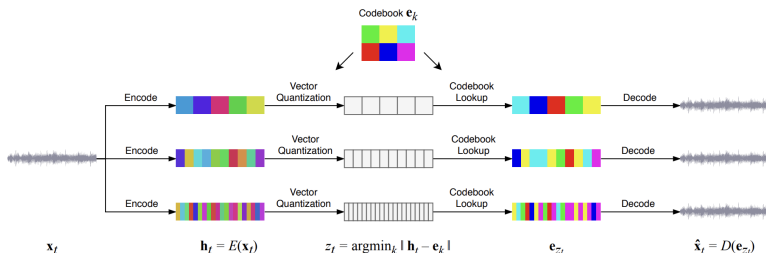


MuseNet (OpenAI, 2019)



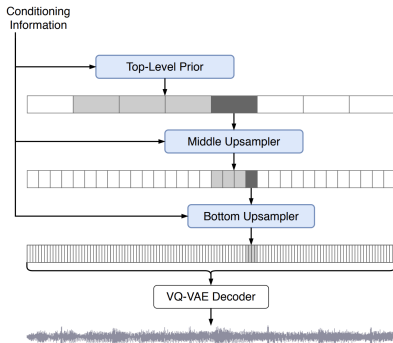
- ▶ MIDI data: ClassicalArchives and BitMidi donations + MAESTRO
- ▶ Task: Learn to predict the next token (given a set of notes, we ask it to predict the upcoming note)
- ▶ Model: 72-layer Sparse Transformer-like network with 24 attention heads with context of **4096 tokens** (Trained like GPT-2)

Jukebox (OpenAI, 2020)



- ▶ 3 VQ-VAE with different temporal resolutions
- ▶ Add additional STFT loss to VQ-VAE standard loss
- ▶ After training VQ-VAE, we need to learn a prior over the compressed space to generate samples
- ▶ $\Rightarrow$ Add **Top-prior** (Transformers with sparse attention) and **Upsamplers models** (WaveNet + upsampling conv + layer norm)

Jukebox: sampling



- ▶ **Conditioning information:** genre, artist, timing, and lyrics
- ▶ **Sampling:** Top-Level Prior → Middle Upsampler → Bottom Upsampler → bottom VQ-VAE decoder
- ▶ Upsamplers are trained on tokens from Bottom and Middle VQ-VAE encoder tokens
- ▶ Top prior is trained on top tokens from VQ-VAE

Music Generation Models: Codecs and Diffusion boosts industry from 2022

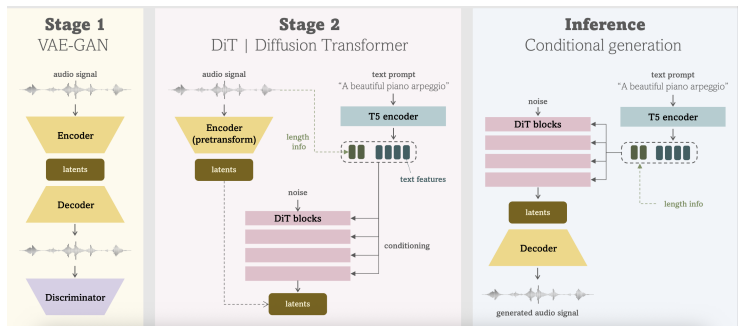
Model Name	Base Architecture	Dataset Used	Data Representation	Loss Function	Year
WaveNet	CNN	VCTK Corpus, YouTube Data	Waveform	L1 Loss	2016
BachBot	LSTM	Bach Chorale Dataset	Symbolic Data	Cross-Entropy Loss	2016
DCGAN	CNN	Lakh MIDI Dataset (LMD)	Audio Waveform	Binary Cross-Entropy Loss	2016
DeepBach	LSTM	Bach Chorale Dataset	MIDI File	Cross-Entropy Loss	2017
MuseGAN	GAN	Lakh MIDI Dataset (LMD)	Multi-track MIDI	Binary Cross-Entropy Loss	2018
MIDI-VAE	VAE	MIDI files (Classic, Jazz, Pop, Bach, Mozart)	Pitch roll, Velocity roll, Instrument roll	Cross-Entropy, MSE, KL Divergence	2018
Music Transformer	Transformer	Lakh MIDI Dataset (LMD)	MIDI File	Cross-Entropy Loss	2019
WaveGAN	GAN	Speech Commands, AudioSet	Audio Waveform	GAN Loss (Wasserstein Distance)	2019
Jukebox	VQ-VAE + Autoregressive	1.2 million songs (LyricWiki)	Audio Waveform	Reconstruction Loss, Perceptual Loss	2019
MelGAN	GAN-based	VCTK, LJSpeech	Audio Waveform	GAN Loss (Multi-Scale Discriminator)	2019
Pop Music Transformer	Transformer-XL	Custom Dataset (Pop piano music)	REMI (Rhythm-Event-Metric Information)	Cross-Entropy Loss	2020
DiffWave	Diffusion Model	VCTK, LJSpeech	Waveform	L1 loss, GAN loss	2020
Riffusion	Diffusion + CLIP	Large-Scale Popular Music Dataset (Custom)	Spectrogram Image	Diffusion Loss, Reconstruction Loss	2022
MusicLM	Transformer + AudioLDM	Free Music Archive (FMA)	Audio Waveform	Cross-Entropy Loss, Contrastive Loss	2023
MusicGen	Transformer	Shutterstock, Pond5	Audio Waveform	Cross-Entropy Loss, Perceptual Loss	2023
Music ControlNet	Diffusion Model	MusicCaps (1800 hours)	Audio Waveform	Diffusion Loss	2023
Mousai	Diffusion Model	Mousai-2023	Mel-spectrogram	Spectral loss, GAN loss	2023
MeLoDy	LM-guided Diffusion	257k hours of non-vocal music	Audio Waveform	Cross-Entropy Loss, Diffusion Loss	2023
MuseCoco	GAN-based	Multiple MIDI datasets including Lakh MIDI and MetaMIDI	Multi-track MIDI	Binary Cross-Entropy Loss	2023
Noise2Music	Diffusion Model	MusicCaps, MTAT, Audioset	Audio Waveform	Diffusion Loss	2023

Chen et al., Applications and Advances of Artificial Intelligence in Music Generation: A Review, 2024

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Stable Audio Open (Stability AI, 2024)



Stable diffusion, but for audio. Problems with raw diffusion:

- ▶ **Short context:** weak for long full-song structure
- ▶ **Limited composition control:** weaker melody, harmony, motif, arrangement control
- ▶ **Slower sampling:** slower than AR/transformer based models
- ▶ **General audio focus:** built for broad audio, less specialized for music

Riffusion (2022): How to make smooth transition?

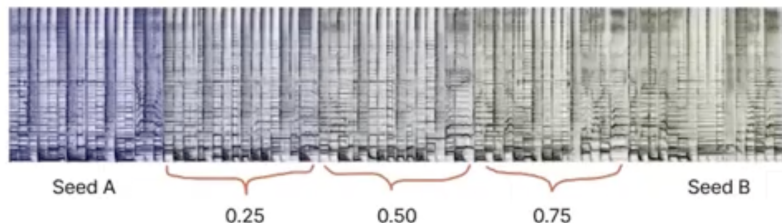
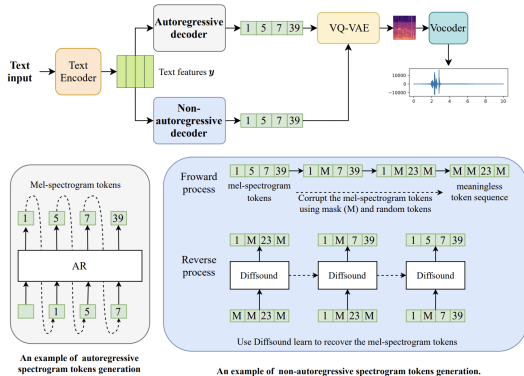


Figure: Latent space interpolation between two seeds of the same prompt

- ▶ Fine-tuned stable diffusion to generate images of spectrograms
- ▶ Combine short clips by sampling the latent space between a prompt with two different seeds, or two different prompts with the same seed
- ▶ laying the sequence is much smoother than just playing the two endpoints
- ▶ Interpolated clips are often diverse and have their own riffs and motifs come and go

Diffsound (Tencent AI Lab, 2022)

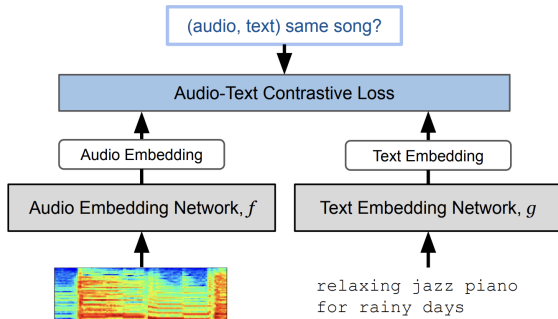


- ▶ Text encoder (CLIP) + Diffusion Decoder + VQ-VAE + Vocoder (MelGAN)
- ▶ Used CLIP because it has better semantic representations
- ▶ Shows that diffusion is better than AR in token decoder

Yang et al., *Diffsound: Discrete Diffusion Model for Text-to-sound Generation*, IEEE, 2023

MuLan: text & music audio joint embedding model

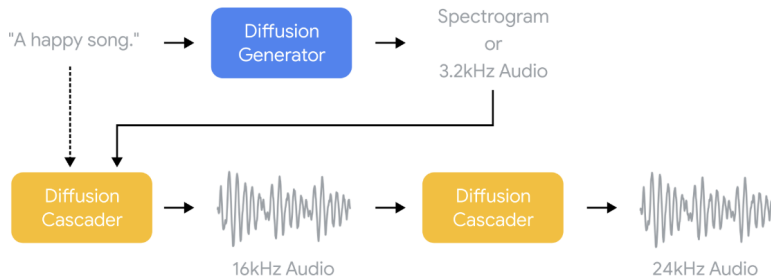
(Google, 2022)



- ▶ **Training dataset:** (audio, text) pairs (370K hours of music videos + text annotation)
- ▶ **Text Embedding:** BERT base-uncased, **Audio Embedding:** Resnet-50/Vision Transformer (ViT) (12 Transformer blocks)
- ▶ **Minimize cross-modal** extension of the **InfoNCE loss**

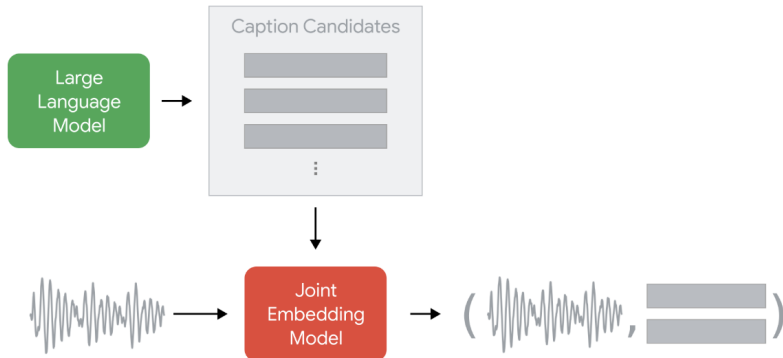
Huang, MuLan: A Joint Embedding of Music Audio and Natural Language, Google, 2022

Noise2Music (Google, 2023)



- ▶ 3 diffusion models (1D Efficient U-Net): **Generator** (724M), **Cascader 487M** (low-fidelity audio generator based on text and Generator representation), **light-weight Cascader 81M** (16kHz waveform $\rightarrow$ 24kHz)
- ▶ Text conditioning: via cross attention

Noise2Music: how to get more data

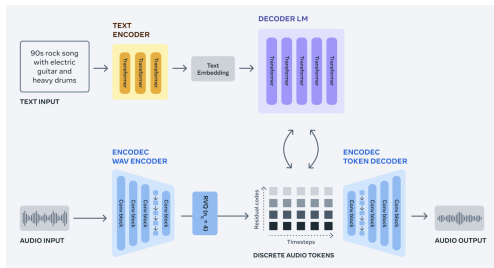


- ▶ Pseudo-labeling for Training Data Generation
- ▶ Joint embeddings: **MuLan**; LLM: **LaMDA**
- ▶ Use MuLan as a zero-shot music classifier to assign captions from LaMDA to unlabeled audio clips

Huang, Noise2Music: Text-conditioned Music Generation with Diffusion Models, Google, 2023

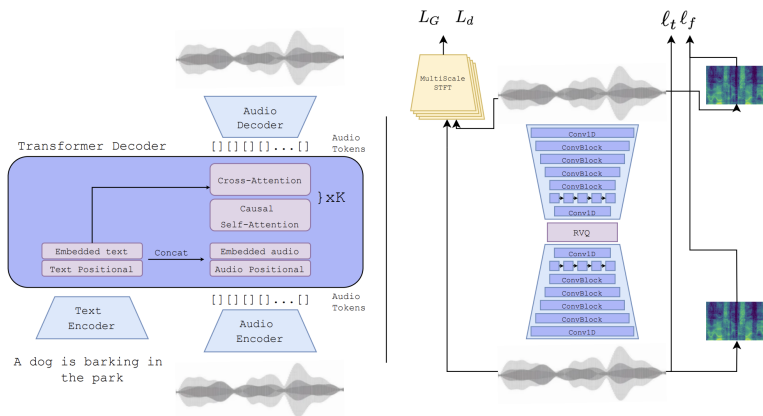
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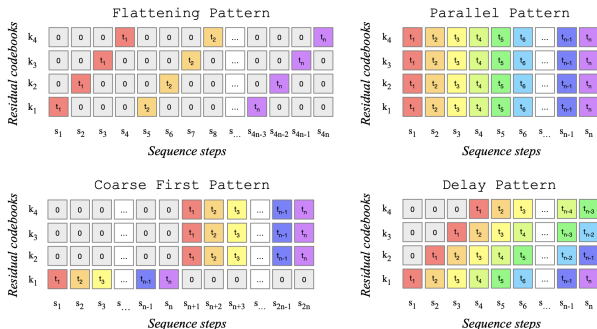
- ▶ **MusicGen**: controllable text-to-music model
- ▶ **AudioGen**: text-to-sound model
- ▶ **EnCodec**: high fidelity neural audio codec
- ▶ **Multi Band Diffusion**: EnCodec + diffusion
- ▶ **MAGNeT**: non-autoregressive text-to-music, text-to-sound
- ▶ **AudioSeal**: audio watermarking
- ▶ **MusicGen Style**: text-and-style-to-music model
- ▶ **JASCO**: text-to-music model conditioned on chords, melodies and drum tracks

AudioGen (Meta AI, 2022)



- ▶ Audio representation: **Encodec (4 RVQ)**
- ▶ Text encoder: **T5**
- ▶ Combine audio and text tokens using **Transformer decoder** (not a cascade system: only 1 Transformer)

MusicGen (Meta AI, 2023)

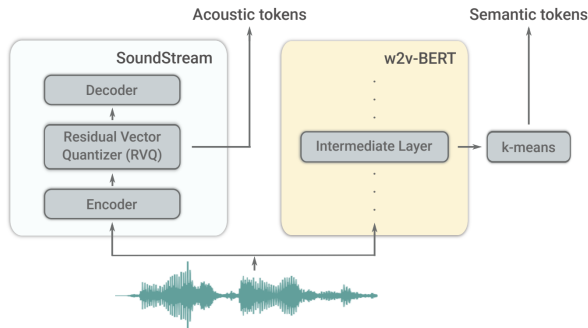


- ▶ AudioGen based: Single stage AR Transformer trained over a 32kHz EnCodec tokenizer with 4 codebooks sampled at 50 Hz
- ▶ Conditioning: on text (T5/FLAN/CLAP) + Melody (chromagram: Visual pitch energy graph)
- ▶ Codebooks sample: delay pattern → predict them in parallel

MusicGen results

MODEL	MUSICCAPS Test Set				
	FAD _{vgg} ↓	KL ↓	CLAP _{scr} ↑	OVL. ↑	REL. ↑
Riffusion	14.8	2.06	0.19	79.31±1.37	74.20±2.17
Mousai	7.5	1.59	0.23	76.11±1.56	77.35±1.72
MusicLM	4.0	-	-	80.51±1.07	82.35±1.36
Noise2Music	2.1	-	-	-	-
MUSICGEN w.o melody (300M)	3.1	1.28	0.31	78.43±1.30	81.11±1.31
MUSICGEN w.o melody (1.5B)	3.4	1.23	0.32	80.74±1.17	83.70 ±1.21
MUSICGEN w.o melody (3.3B)	3.8	1.22	0.31	84.81 ±0.95	82.47±1.25
MUSICGEN w. random melody (1.5B)	5.0	1.31	0.28	81.30±1.29	81.98±1.79

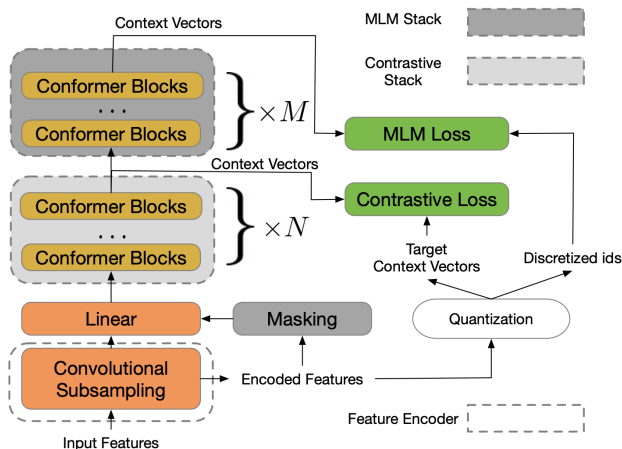
AudioLM (Google Research, 2022)



- ▶ **SoundStream**: 24 kHz audio to 50 Hz embeddings (Autoencoder codec with RVQ)
- ▶ **w2v-BERT** (wav2vec2 + MLM loss): extract embeddings from the 7th layer and quantize them using the centroids of k-means over the embeddings

Borsos, Marinier et al., AudioLM: a Language Modeling Approach to Audio Generation, IEEE/ACM, 2022

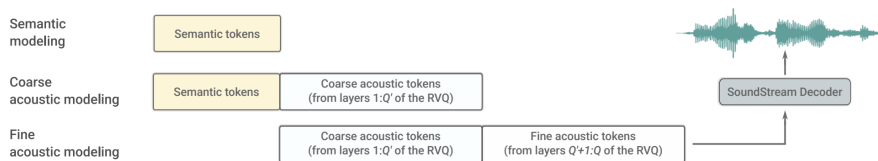
w2v-BERT



Similar to w2v-Conformer, but with additional MLM training loss

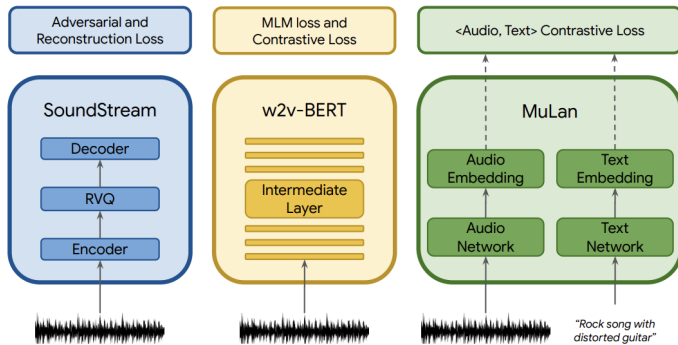
Chung et al., W2v-BERT: Combining Contrastive Learning and Masked Language Modeling for Self-Supervised Speech Pre-Training, MIT+Google, 2021

AudioLM: decoding



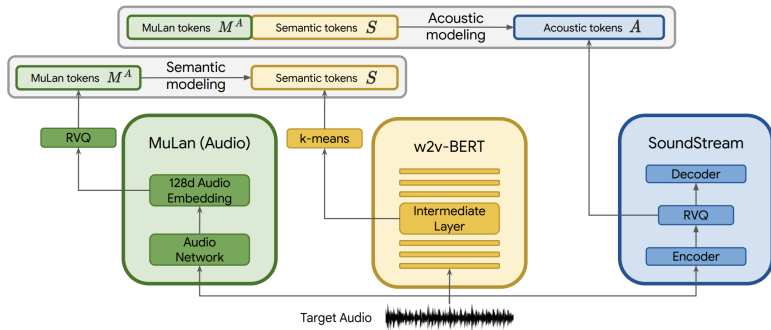
Acoustic and Fine acoustic modeling: decoder-only Transformers trained for predicting next tokens

MusicLM (Google Research, 2023)



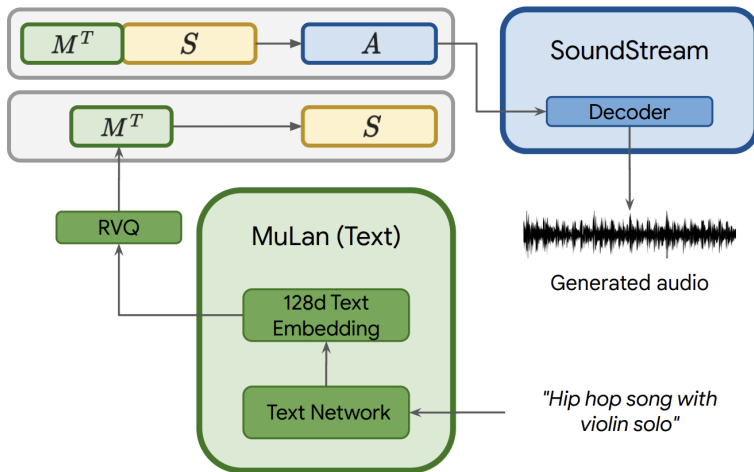
- ▶ Based on AudioLM: (SoundStream + w2v-BERT) + **MuLan**
- ▶ MuLan tokens: joint embeddings for music & text
- ▶ Metrics Frechet Audio Distance (FAD), KL

MusicLM: train

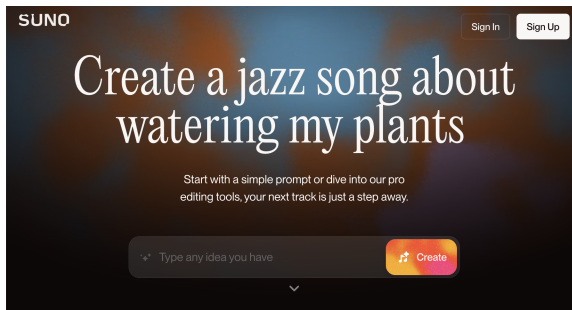


- ▶ Pretrained: SoundStream, w2v-BERT, MuLan (frozen)
- ▶ **Train** together with **only audio input data**
- ▶ Trains only: 2 Decoder-only Transformers: Semantic LM + Acoustic LM

MusicLM: sampling

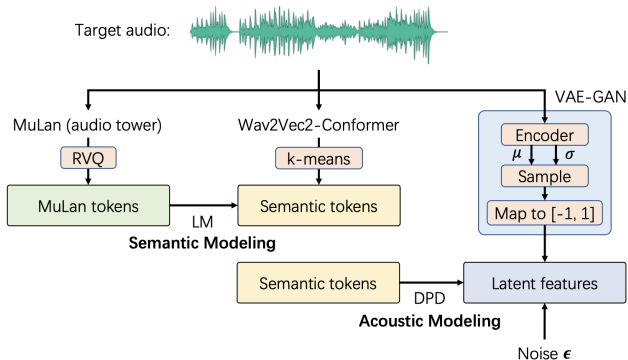


Bark: open-source model from Suno.ai (2023)



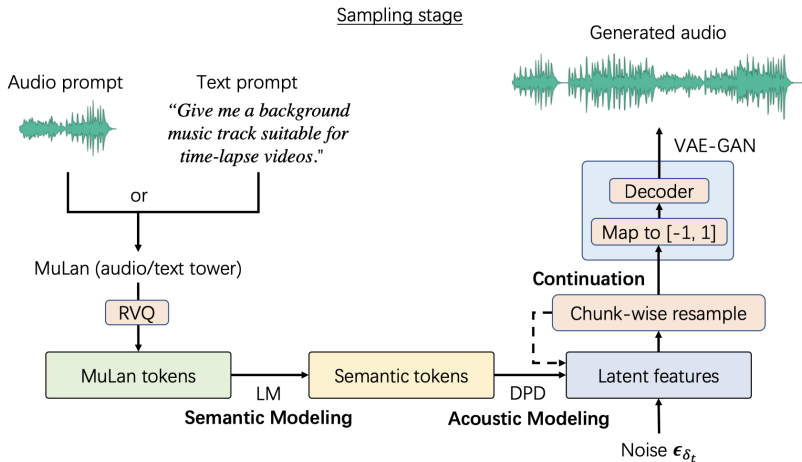
- ▶ Generates speech, audio, music
- ▶ Architecturally very similar to Google's AudioLM + VALL-E
- ▶ Bark is a series of 3 transformer GPT-base models:
 1. **Text to semantic tokens** (80 M)
 2. **Semantic to coarse tokens** (80 M, coarse tokens: first 2 codebooks of the EnCodec)
 3. **Coarse to fine tokens** (80 M, **Input**: the first 2 codebooks from **EnCodec**, **Output**: 8 codebooks from EnCodec)

MeLoDy training (**M** for music; **L** for LM; **D** for diffusion) (ByteDance, 2023)



- ▶ LM-guided Diffusion
- ▶ MuLan + Wav2Vec2-Conformer + VAE-GAN + Dual-path diffusion (DPD)
- ▶ Training data: 257k hours of non-vocal music

MeLoDy: sampling



MeLoDy: comparison with SOTA models

Model	Prompts	Training Data	AC	FR	VT	MP
Moûsai [23]	Text	2.5k hours of music	✓	✓	✗	✗
MusicLM [5]	Text, Melody	280k hours of music	✓	✗	✓	✗
Noise2Music [6]	Text	340k hours of music	✗	✗	✓	✗
MeLoDy (Ours)	Text, Audio	257k hours of music ¹	✓	✓	✓	✓

- ▶ **AC**: audio continuation is supported
- ▶ **FR**: the sampling is faster than real-time on a V100 GPU
- ▶ **VT**: the model has been tested and demonstrated using various types of text prompts including instruments, genres, and long-form rich descriptions
- ▶ **MP**: the evaluation was done by music producers

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Instrumental Leaderboard (Artificial Analysis, 2026)

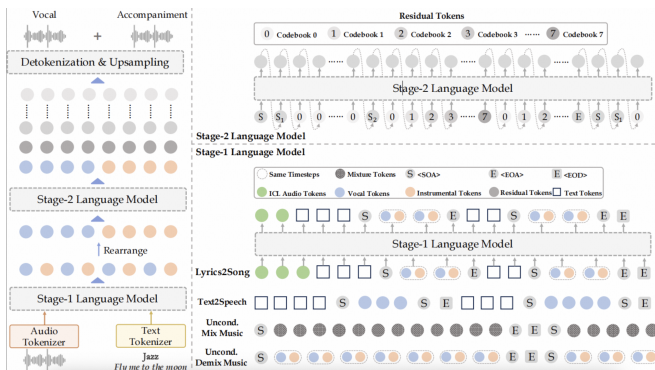
↑↓	Range ⓘ	Creator ↑↓	Model ↑↓	ELO ↑↓	95% CI	Samples ↑↓
1	1	 Mureka	Mureka V8	1,182	-13/13	3,452
2	2	 Suno	Suno V5	1,164	-14/14	2,722
3	3	 Google	Lyria 3 Pro	1,128	-14/14	2,636
4	4	 Suno	Suno V4.5	1,106	-13/13	2,759
5	5-6	 II ElevenLabs	Eleven Music	1,055	-13/13	2,732
6	5-7	 MiniMax	MiniMax Music 2.5+	1,051	-14/14	2,640
7	6-8	 Producer.ai	FUZZ-11 Pro	1,039	-13/13	2,464
8	7-8	 Producer.ai	FUZZ-2.0 Raw	1,036	-13/13	3,074
9	9	 Producer.ai	FUZZ-2.0	1,023	-13/13	2,935
10	10-9	 Google	Lyria 2	1,000	0/0	2,631

Leaders: Mureka (China), Suno, ElevenLabs, Google, ProducerAI (ex-Riffusion, now Google), MiniMax, Udio

Vocals Leaderboard (Artificial Analysis, 2026)

↑↓	Range 	Creator ↑↓	Model ↑↓	ELO ↑↓	95% CI	Samples ↑↓
1	1	 Mureka	Mureka V8	1,144	-11/11	3,431
2	2-5	 Google	Lyria 3 Pro	1,090	-12/12	2,493
3	2-5	 Suno	Suno V5	1,083	-11/11	3,046
4	2-5	 MiniMax	MiniMax Music 2.5+	1,079	-12/12	2,717
5	3-5	 Suno	Suno V4.5	1,079	-11/11	4,000
6	6	 Producer.ai	FUZZ-2.0	1,058	-11/11	4,033
7	7-8	 ElevenLabs	Eleven Music	1,011	-11/11	3,831
8	7-8	 Producer.ai	FUZZ-2.0 Raw	1,011	-11/11	4,490
9	9-8	 Producer.ai	FUZZ-1.1 Pro	1,000	0/0	3,958
10	10	 Udio	Udio v1.5 Allegro	946	-11/11	4,295

Lets generate full song from lyrics: YuE (2025)



- ▶ Audio, Text tokenizers + 2-Stage LM + Detokenizer
- ▶ 2 token streams simultaneously: vocals + instrumental

$$\text{ICL Tokens} = \left[\underbrace{T_{\text{reference}}}_{\text{lyrics}}, \underbrace{A_{\text{reference}}}_{\text{audio tokens}}, \underbrace{S_{\text{delim}}}_{\text{separator}}, \underbrace{T_{\text{target}}}_{\text{lyrics / prompt tokens}} \right]$$

Yuan et al., YuE: Scaling Open Foundation Models for Long-Form Music Generation, HKUST, 2025

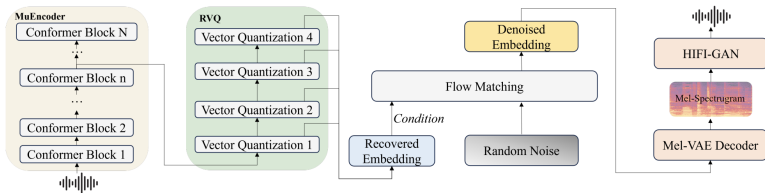
YuE: architecture

1. **Text Tokenizer**: LLaMA-2 SentencePiece-BPE (32k vocab)
2. **Audio tokenizer**: X-Codec (semantic-acoustic fused RVQ-VAE, HuBERT-base, 100 M)
3. **Stage-1 LM** (“MuLM”): LLaMA-2-style decoder-only Transformer (0.5 B $\rightarrow$ 7 B)
4. **Stage-2 LM** (“Res-LM”): Llama2-1B (predicts 1-7 codebooks from codebook 0)
5. **Detokenizer**: X-Codec decoder
6. **Light vocoder-up sampler**: HiFi-GAN V2 (12 M)

YuE: results

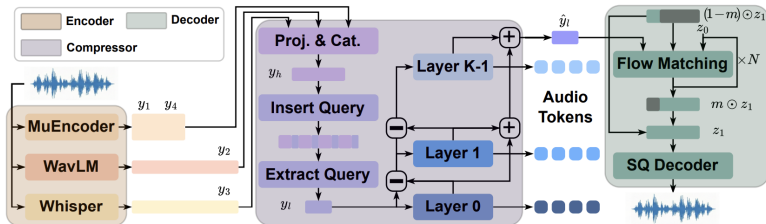
Metric	Distrib. Match		Content Based				Alignment	
	KL↓	FAD↓	CE↑	CU↑	PC↑	PQ↑	CLAP↑	CLaMP 3↑
Hailuo	0.756	2.080	7.350	7.737	6.793	8.132	0.265	0.106
SunoV4	0.620	1.544	7.474	7.813	6.601	8.120	0.265	0.160
Tiangong	0.708	2.547	7.421	7.766	6.060	8.220	0.244	0.114
Udio	0.503	1.222	7.112	7.520	6.626	7.803	0.310	0.156
YuE	0.372	1.624	7.115	7.543	6.280	7.894	0.118	0.240

Let's make codec specially for music: MuCodec (Tencent, 2025)



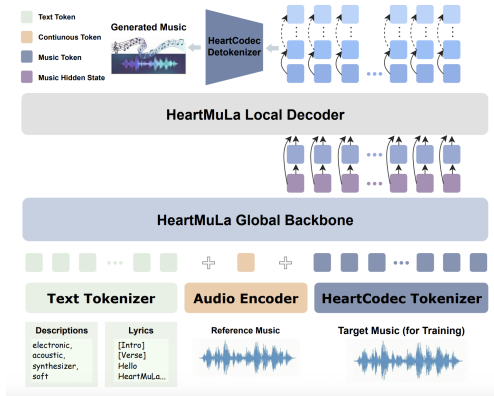
- ▶ RVQ codec
- ▶ Encoder: 13 Conformer blocks
- ▶ Decoder: Diffusion Transformer (DiT) Flow-Matching + Mel-VAE + HiFi-GAN

HeartCodec (from HeartMuLa)



- ▶ Whisper: phonetic / vocal cues
- ▶ WavLM: acoustic detail
- ▶ MuEncoder: music semantics (from MuCodec)
- ▶ Decoder: conv $\rightarrow$ flow matching

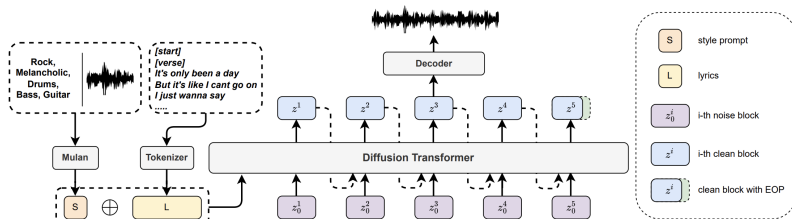
Let's make everything from scratch for music: HeartMuLa (2026)



- ▶ Text encoder: Llama-3.2 tokenizer
- ▶ Audio encoder: HeartCLAP (audio-text alignment like MuLan)
- ▶ Use special codec for music: HeartCodec

Yang et al., *HeartMuLa: A Family of Open Sourced Music Foundation Models*, HeartMuLa, 2026

Let's generate full song from lyrics with diffusion: DiffRhythm 2 (2026)



- ▶ Diffusion: flow matching generates VAE latents block by block
- ▶ Decoder: Music VAE (low frame rate 5 Hz to tackle long sequences)
- ▶ Music VAE: Encoder (Stable Audio 2 VAE) + transformer block + Decoder (BigVGAN)
- ▶ Lyric alignment: blockwise generation
- ▶ Train with DPO, not RLHF

Jiang et al., *DiffRhythm 2: Efficient and High Fidelity Song Generation via Block Flow Matching*, 2026

Other significant recent works: full song generation trend (lyric alignment)

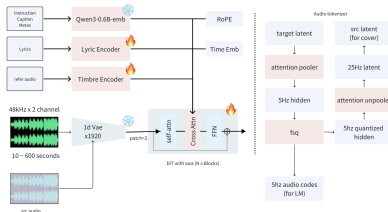


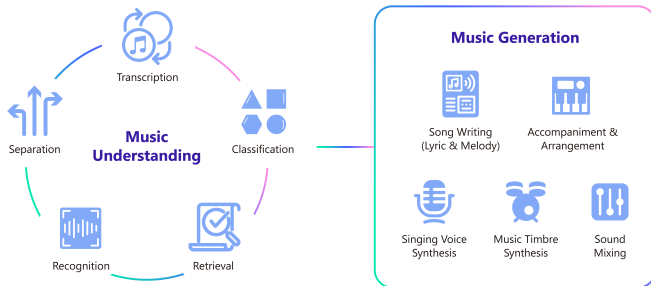
Figure: ACE-Step 1.5 pipeline

- ▶ **SongGen (2025)**: introduces interleaved prediction across tracks
- ▶ **LeVo (Tencent, 2025)**: first generate a mixed track, then use it to guide separate vocal and accompaniment synthesis
- ▶ **ACE-Step (ACE Studio, 2025)**: apply diffusion similar to DiffRhythm
- ▶ **ACE-Step 1.5 (ACE Studio, 2025)**: planning (LM) + diffusion (flow matching)

ACE-Step 1.5 results

Model	AudioBox $\uparrow$				SongEval $\uparrow$					Style Align $\uparrow$	Lyric Align $\uparrow$
	CE	CU	PC	PQ	Coh.	Mus.	Mem.	Cla.	Nat.		
Udio-v1.5	7.45	7.65	6.15	8.03	4.15	3.96	4.09	3.93	3.86	34.9	24.8
Suno-v4.5	7.63	7.85	6.22	8.25	4.64	4.51	4.63	4.53	4.49	40.5	<u>32.7</u>
Suno-v5	<u>7.69</u>	7.87	6.51	8.29	4.72	<u>4.62</u>	<u>4.71</u>	<u>4.63</u>	<u>4.56</u>	46.8	34.2
Mureka-V7.6	7.44	7.71	6.35	8.13	4.43	4.29	4.35	4.29	4.21	36.2	22.4
MinMax-2.0	7.71	<u>7.95</u>	6.42	8.38	4.61	4.51	4.59	4.50	4.41	<u>43.1</u>	29.5
Yue [18]	6.58	7.29	4.95	7.39	3.01	2.80	2.85	2.79	2.82	26.8	-4.6
ACE-Step 1.0	7.22	7.52	<u>6.50</u>	7.76	3.99	3.73	3.85	3.78	3.68	28.5	0.9
LeVo [19]	7.61	7.78	5.92	8.31	3.55	3.35	3.32	3.31	3.20	29.4	-1.2
DiffRhythm 2 [20]	7.25	7.61	6.33	7.99	3.99	3.79	3.97	3.82	3.66	32.1	3.8
HeartMuLa [21]	7.66	7.89	6.15	8.25	<u>4.68</u>	4.55	4.69	4.55	4.45	31.7	28.6
ACE-Step 1.5	7.42	8.09	6.47	<u>8.35</u>	4.72	4.67	4.72	4.66	4.59	39.1	26.3

Muzic (Microsoft Research Asia)



1. Music Understanding

- ▶ Symbolic Music Understanding: MusicBERT
- ▶ Automatic Lyrics Transcription: PDAugment
- ▶ Contrastive Language-Music Pre-training: CLaMP

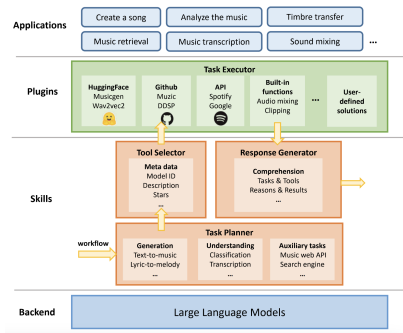
2. Music Generation

3. AI Agent: MusicAgent

Muzic: Music Generation models

- ▶ Song Writing
 - ▶ Lyric-to-Melody and Melody-to-Lyric: SongMASS
 - ▶ Lyric Generation: DeepRapper
 - ▶ Lyric-to-Melody Generation: TeleMelody, ReLyMe, Re-creation of Creations (ROC)
- ▶ Music Form Generation: MeloForm
- ▶ Long/Short Structure Modeling: **Museformer**
- ▶ Accompaniment Generation: PopMAG
- ▶ Any Track Music Generation: GETMusic
- ▶ Text-to-Music Generation: **MuseCoco**
- ▶ Singing Voice Synthesis: **HiFiSinger**

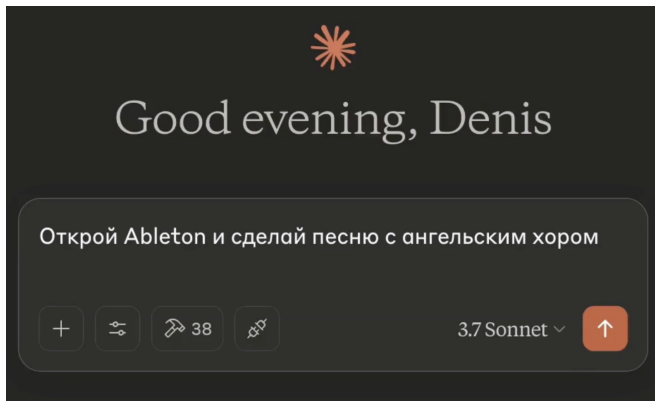
LLM Agents + Audio: MusicAgent by Microsoft Research Asia



- ▶ **4 core components:** the task planner (LLM), tool selector (LLM), task executor, and response generator (LLM).
- ▶ User request → decomposes it into subtasks → selects the most suitable tool for each task (and process each) → organizes the output

Yu et al., MusicAgent: An AI Agent for Music Understanding and Generation with Large Language Models, Microsoft Research Asia, 2023

LLM + MCP + Ableton: Vibe Music Generation



Generate music in professional music applications using LLM agents with MCP servers: example with Claude